Sri Lanka Institute of Information Technology

User Acceptance Testing Report

For

IoT-Enabled Smart Bus Route Monitoring and Revenue Management System

Project ID: R26-ISE-004

B.Sc. (Hons) in Information Technology Specializing in Information Systems Engineering

Prepared by:

Registration No.	Student Name
IT22228202	W S H R Gunasekara
IT22121042	Ahamed M
IT22264152	Sachintha H D D
IT22925268	Aththanayaka N D

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List Of Abbreviations

Abbreviation	Description
UAT	User Acceptance Testing
IoT	Internet of Things
GPS	Global Positioning System
IMU	Inertial Measurement Unit
MQTT	Message Queuing Telemetry Transport
TLS	Transport Layer Security
JWT	JSON Web Token
API	Application Programming Interface
REST	Representational State Transfer
RBAC	Role-Based Access Control
NTC	National Transport Commission
WPPTA	Western Province Passenger Transport Authority
LKR	Sri Lankan Rupee
UI	User Interface
DB	Database

1. Introduction

This report presents the User Acceptance Testing (UAT) procedures and results for the BusSync platform, an IoT-based framework that replaces flat permit fees with verified kilometres for private bus services in Sri Lanka. Before final deployment, UAT confirms whether the platform meets the expectations of its users: transport authority officers, bus owners, and bus drivers. Testing covers the four individual components of the platform: bus tracking, route violation detection and halt detection; policy-driven bus scheduling and time-slot management; distance-based revenue management; and real-time driver behavior telematics with the stakeholder portals. It also covers authentication, access control, and the end-to-end flow joining the components. Every engine is algorithm-based and rule-based; therefore, UAT evaluates whether each decision, calculated value, and alert matches the expected outcome. No model training or prediction-accuracy measure is involved.

1.1. Project Overview

BusSync is an IoT-enabled intelligent transportation and revenue governance framework. Each bus carries an ESP32 edge unit with a NEO-6M GPS module, an MPU-6050 six-axis IMU, and a SIM800L modem. The unit samples GPS at 1 Hz and the IMU at 50 Hz and sends JSON telemetry over TLS-encrypted MQTT to Microsoft Azure IoT Hub. A Spring Boot backend with PostgreSQL and PostGIS processes telemetry. Users access the platform through a React Native mobile application for drivers and bus owners and a React.js web portal built with Ant Design for the transport authority.

- Bus Tracking, Route Violation Detection and Halt Detection (Component 1): receives and verifies telemetry, projects positions onto the digitized route using point-to-segment map-matching against a 25 m corridor, flags a route deviation when a bus remains outside the corridor for more than 30 seconds, detects halts, and records verified kilometers.
- Policy-Driven Bus Scheduling and Time-Slot Management Engine (Component 2): generates the monthly base schedule from the route timetable and handles authority-approved disruption events through reallocation or cancellation with a stored reason.
- Distance-Based Revenue Management System (Component 3): calculates monthly permit liability from verified kilometers at LKR 12.00 per km, applying multipliers of

1.30 (Peak), 1.00 (Normal), and 0.70 (Night), and deducts a night-service incentive of LKR 25.00 per verified night kilometer.

- Real-Time Driver Behavior Telematics and Stakeholder Portals (Component 4): detects over speeding, aggressive braking, aggressive turning, and abnormal slow movement from GPS speed and IMU data, alerts the driver in the cab, updates the driver safety score, and presents results in the mobile application and web portal.

The components form a closed loop. Component 2 supplies the scheduled turn and time slot; Component 1 supplies verified distance and route compliance; Component 3 uses this data to calculate the monthly bill; and Component 4 reports driving behavior to the driver, owner, and authority.

1.2. Scope

The UAT covers functions accessed through the authority web portal, owner and driver mobile application, and ESP32 edge unit. It includes login and role-based access control; telemetry ingestion, live tracking, route deviation and halt detection, verified distance, signal and network loss handling, and compliance auditing (Component 1); monthly base scheduling, time-slot management, disruption requests and review, reallocation, cancellation, and updated schedules (Component 2); monthly bills, night incentives, settlement, and role-based billing access (Component 3); in-cab alerts, safety scores, and portal safety views (Component 4); and four end-to-end flows across components. The prototype is tested on Route 138 (Pettah–Homagama) and Route 177 (Kollupitiya–Kaduwela). Rural unpaved routes, expressway coaches, OBD-II engine diagnostics, automated fare collection, and large-scale production load testing are outside the scope.

1.3. Objectives

The aim of UAT is to confirm that the web and mobile applications and IoT layer are ready for deployment and meet the needs of authority officers, bus owners, and drivers. The specific objectives are:

- Verify functionality through the interfaces used by stakeholders across all four components.
- Verify algorithm correctness for map-matching and deviation detection, scheduling and reallocation, tariff and incentive calculations, and driving-behavior rules, including boundary cases.
- Verify the IoT chain from ESP32 telemetry through MQTT/TLS to the backend, including GPS and network loss handling and its effects on tracking, billing, and alerts.
- Verify role-based usability and ensure each stakeholder sees only the information and controls permitted by their role.
- Verify integration from monthly scheduling to trips, verified kilometres, billing, and driver alerts.
- Verify JWT authentication, role-based access control, TLS enforcement, and the recording and resolution of reliability defects.

2. UAT Prerequisites

Before UAT begins, the platform must be deployed in a controlled test environment where all four components are integrated and operational. Testers require valid credentials for each role, a briefing on the workflows under test, and documented acceptance criteria.

2.1. UAT Entry Criteria

UAT begins once the following conditions have been met:

- Stable integration: the ESP32 unit, Azure IoT Hub, Spring Boot backend, PostgreSQL/PostGIS database, web portal, and mobile application exchange data correctly.
- Defect clearance: critical and high-priority defects from unit, integration, and system testing have been fixed and recorded.
- Documentation approval: test scenarios, test cases, and acceptance criteria are prepared and reviewed.
- Test data: routes, buses, drivers, owners, timetables, tariffs, and thresholds in Table 4 are loaded.
- Hardware readiness: at least one calibrated ESP32 test unit with GPS and IMU is mounted in a test vehicle.
- Security readiness: HTTPS, TLS 1.3 for MQTT, JWT authentication, and role-based access control are active; the environment is isolated from production data.
- User readiness: testers understand the authority officer, bus owner, and driver workflows.

2.2. Test Environment

The test environment reproduces the deployed architecture using local and cloud infrastructure.

i. Hardware and Software Configurations:

Table 1: Hardware and Software Configuration

Component	Specification
Edge Device	ESP32-WROOM-32 with NEO-6M GPS, MPU-6050 six-axis IMU, and SIM800L GPRS modem
Cloud IoT Gateway	Microsoft Azure IoT Hub; MQTT over TLS 1.3
Backend	Java 17; Spring Boot 3.x with Spring Security and JWT
Database	PostgreSQL 15 with PostGIS 3.4
Mobile Frontend	React Native (owner and driver application)
Web Frontend	React.js with Ant Design and Leaflet maps (authority portal)
Decision Logic	Rule-based and algorithm-based; no machine-learning model
Testing Tools	Postman, JUnit, Mockito
Version Control	GitHub
Cloud Platform	Microsoft Azure App Service

ii. Test Interfaces and Access Details

Table 2: Test interfaces and Access Details

Interface	Used by	Access Method
Authority web portal	Authority officer	Browser login with authority role
Owner mobile application	Bus owner	Android application login with owner role
Driver mobile application	Driver	Android application login with driver role
REST API	Tester, through Postman	Endpoints under /api with a Bearer JWT
Azure IoT Hub	ESP32 test unit	MQTT over TLS with device credentials
PostgreSQL and PostGIS	Tester (read-only checks)	Database client with a read-only account

iii. Test Device Configuration

Table 3: Device Configuration

Device	Purpose
ESP32 test unit mounted on the test vehicle	Live telemetry, route deviation, halt, signal loss, and driving-maneuver tests
Reference RTK-GPS receiver	Ground truth for position, speed, and distance
Android emulator or physical Android smartphone	Owner and driver applications: schedule, trip start, requests, alerts, and bills
Desktop web browser	Authority portal: schedule generation, request review, live map, compliance audit, and billing

iv. Reference Test Data

Table 4: Reference Test Data

Item	Reference
Routes	Route 138 (Pettah–Homagama, 24 km) and Route 177 (Kollupitiya–Kaduwela, 18 km)
Buses and drivers	10 active buses (B01–B10) with regular drivers (D01–D10)
Timetable and month	20 trips a day; September 2026 (30 days) gives 600 assignments and 60 trips per bus
Corridor and deviation settings	25 m corridor and 30 seconds of persistence
Scheduling constraints	At least 15 minutes between arrival and next departure of the same bus
Time slots	Morning Peak 06:30–09:00; Evening Peak 16:30–19:00; Normal 09:00–16:30 and 19:00–21:00; Night 21:00–05:00
Tariff settings	Base rate LKR 12.00/km; multipliers 1.30 (Peak), 1.00 (Normal), 0.70 (Night); night incentive LKR 25.00/km
Safety thresholds	Speed above 60 km/h for 5 s; deceleration beyond -3.5 m/s²; yaw rate above 0.45 rad/s at over 25 km/h; speed below 10 km/h for 60 s
Test accounts	One authority officer, owner of Bus B01, and Drivers D01–D03

2.3. Roles and Responsibilities

UAT is a team activity. Each member tests the component they developed through the roles that use it, while the team jointly tests shared flows.

- Test execution: run owned-component cases through the portal, mobile application, REST API, and ESP32 unit, and record actual results.
- Requirement compliance: confirm compliance with functional, non-functional, and safety requirements.
- Issue reporting: record each defect against its test case ID for correction and retesting.
- Test data management: prepare routes, buses, drivers, timetables, trips, and sensor data for each expected outcome.
- Documentation: maintain test logs, results, and defect records.

Collaboration approach: cross-component cases are executed jointly; defects are discussed and assigned for correction; and individual component testing is followed by group integration testing.

Individual Component Ownership

Registration No.	Student Name	Contribution
IT22228208	Gunasekara W. S. H. R.	Distance-Based Revenue Management Testing
IT22121042	Ahamed M.	Bus Tracking, Route Violation Detection and Halt Detection Testing
IT22264152	Sachintha H. D. D.	Policy-Driven Bus Scheduling and Time-Slot Management Testing
IT22925268	Aththanayaka N. D.	Real-Time Driver Behavior Telematics and Stakeholder Portals Testing

3. Test Scenarios and Test Cases

This section defines the scenarios and test cases used to validate BusSync. Each case records its steps, expected result, actual result, and status. Testing begins with authentication and access control, continues through the components in data-flow order, and concludes with end-to-end integration. A case passes only when the calculated value or decision, stored status, and user-visible result agree with the expected outcome for the prepared test data. Boundary cases are included where thresholds determine outcomes.

3.1. Authentication and Access Control

3.1.1. Test Scenario

This scenario validates how users enter the platform and which functions each role can access. The authority officer logs in to the web portal, while the owner and driver use the mobile application. Tests include valid and invalid credentials, role separation, owner-specific data access, and rejection of REST requests without a valid JWT. The objective is to confirm that only authorized users can access the system and that role permissions are enforced.

3.1.2. Test Cases for Authentication and Access Control

Table 5: Test cases for Authentication & Access Control

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-001	Authority officer logs in with valid credentials	1. Open web portal. 2. Enter valid authority credentials. 3. Click Login.	Officer reaches the authority dashboard.	Officer logged in and reached the dashboard.	Passed
UAT-002	Login with invalid credentials	1. Open web portal. 2. Enter an invalid username or password. 3. Click Login.	Login is denied and an error message appears.	Login denied; error message shown.	Passed
UAT-003	Bus owner logs in to mobile application	1. Open owner application. 2. Enter valid owner credentials. 3. Tap Login.	Owner reaches the owner home screen.	Owner logged in and reached home screen.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-004	Driver logs in to mobile application	1. Open driver application. 2. Enter valid driver credentials. 3. Tap Login.	Driver reaches the driver home screen.	Driver logged in and reached home screen.	Passed
UAT-005	Driver cannot reach authority functions	1. Log in as D01. 2. Try schedule generation, request review, and billing settlement via UI or Postman.	Access is denied for every function.	Access denied for every function.	Passed
UAT-006	Owner sees only the owner's own data	1. Log in as owner of B01. 2. Open bus list, schedule, and bills.	Only the owner's buses and related data are shown.	Only the owner's buses and data shown.	Passed
UAT-007	Request without a valid token is refused	1. Call a REST endpoint without JWT. 2. Repeat with expired JWT.	Both requests return unauthorised responses and no data.	Both requests refused; no data returned.	Passed

3.2. Bus Tracking, Route Violation Detection and Halt Detection

3.2.1. Test Scenario

This scenario validates the platform's spatial evidence layer. The ESP32 unit connects to Azure IoT Hub over MQTT/TLS and sends packets containing device ID, trip ID, GPS position, and IMU values. Tests cover packet validation, live tracking, speed comparison with the reference receiver, corridor compliance, persistent route deviations, verified distance, halt detection, GPS loss, network buffering, and compliance audit output.

3.2.2. Test Cases for Bus Tracking, Route Violation Detection and Halt Detection

Table 7: Test Cases for Bus Tracking, Route Violation Detection and Halt Detection

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-008	ESP32 connects to Azure IoT Hub	Power on test unit; open the device list in authority portal.	Unit connects over MQTT/TLS and appears online.	Unit connected and appeared online.	Passed
UAT-009	Valid telemetry packets are accepted	Start trip as D01; send packets for one minute; inspect logs and records.	Valid packets are acknowledged and stored.	Packets acknowledged and stored.	Passed
UAT-010	Malformed packet is rejected	Send packet without trip ID through Postman; inspect response and storage.	Bad-request response; no record stored.	Request rejected; nothing stored.	Passed
UAT-011	Live position meets latency target	Open Live Map and move test vehicle.	Marker updates with end-to-end latency below 2.0 s.	Latency below 2.0 s observed.	Passed
UAT-012	Displayed speed matches reference	Drive at 30–50 km/h; compare portal and reference receiver.	Difference is within 5 km/h.	Difference within 5 km/h.	Passed
UAT-013	Inside-corridor travel raises no alert	Drive inside 25 m corridor; watch alerts.	No deviation alert; distance accumulates.	No alert; distance accumulated.	Passed
UAT-014	Short corridor exit is ignored	Leave corridor for about 20 s,	No deviation event is flagged.	No deviation event flagged.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
		return, inspect alerts.			
UAT-015	Deviation is flagged after 30 seconds	Stay outside corridor for more than 30 s.	Deviation event and portal alert appear.	Event and alert appeared.	Passed
UAT-016	Deviation alert displays details	Open a deviation alert.	Trip, bus, start time, location, and duration are shown.	All details shown.	Passed
UAT-017	Returning to corridor closes deviation	Return inside corridor; inspect event.	Event closes with duration; distance accumulation resumes.	Event closed; accumulation resumed.	Passed
UAT-018	Verified distance matches reference	Complete 24 km Route 138 trip; compare reference distance.	Difference is within 2%.	Values agreed within 2%.	Passed
UAT-019	Distance outside corridor is not verified	Drive about 1 km in deliberate deviation; inspect summary.	Outside-corridor kilometres are excluded.	Outside-corridor distance excluded.	Passed
UAT-020	Halt on corridor is detected	Stop on active corridor beyond configured threshold.	Halt event with location and start time appears.	Halt event appeared with details.	Passed
UAT-021	Halt closes when bus moves	Move after halt; inspect record.	Halt closes with duration.	Halt closed with duration.	Passed
UAT-022	GPS loss does not cause false penalty	Cover GPS antenna for over 45 s; restore signal; inspect trip.	Segment marked Interpolated; no deviation penalty.	Segment interpolated; no penalty.	Passed
UAT-023	Network loss is buffered and recovered	Disable cellular service for minutes; restore and compare records.	Buffered records arrive in order without loss.	Records recovered in order; no loss observed.	Passed
UAT-024	Compliance audit report is produced	Open audit; select Route 138 and	Report shows total/compliant trips,	Required report fields shown.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
		date range; generate.	violations, and deviation logs.		
UAT-025	Invalid audit date range is refused	Enter start date after end date and generate.	Report is blocked and validation message appears.	Report blocked; message shown.	Passed

3.3. Policy-Driven Bus Scheduling and Time-Slot Management

3.3.1. Test Scenario

The Monthly Base Scheduling scenario validates the planning stage of the engine. The authority officer logs in to the web portal, opens the monthly schedule generator, selects the route, the timetable of 20 trips a day, and the month 2026-09, and triggers generation. The backend goes through every day of the month, takes the trips in departure order, and gives each trip to the eligible bus with the fewest trips so far. The scenario checks that 600 assignments are stored, that the monthly summary shows 60 trips for each of the 10 active buses, that no bus holds overlapping assignments, that the 15-minute turnaround is respected, and that inactive buses receive no trip. It also confirms that the total follows the timetable and the calendar rather than the fleet size, that repeated generation with the same input gives the same schedule, that ties are settled by continuous turn preference and then by bus ID, and that bus owners and drivers cannot reach the generator. Overall, the scenario shows that the monthly base schedule is complete, fair, and free of conflicts.

3.3.2. Test Cases for Bus Monthly Base Scheduling

Table 6: Test Cases for Monthly Base Scheduling

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-026	Generate monthly base schedule with valid input	Select Route 138, 20-trip timetable, September 2026; click Generate.	Schedule generated; summary shows 600 trips.	Schedule generated; 600 trips shown.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-027	Validate required fields	Leave month empty; click Generate.	No schedule; validation message shown.	No schedule; validation shown.	Passed
UAT-028	Assignments follow calendar and timetable	Generate February 2027 schedule with 20 trips/day.	Summary shows 560 assignments.	560 assignments shown.	Passed
UAT-029	Trips shared evenly across active buses	Generate September 2026 schedule; inspect trips per bus.	Each bus has 60 trips, or the smallest possible spread.	Each bus had 60 trips.	Passed
UAT-030	No overlapping assignments	Compare consecutive assignments for each bus.	No overlapping times.	No overlaps found.	Passed
UAT-031	Turnaround constraints respected	Compare arrival and next departure for each bus.	Every gap is at least 15 minutes.	All gaps at least 15 minutes.	Passed
UAT-032	Inactive bus excluded	Mark B05 inactive; regenerate; search schedule.	B05 receives no assignment.	B05 received no assignment.	Passed
UAT-033	Schedule generation is repeatable	Generate twice with identical input; compare assignments.	Both runs produce identical assignments.	Assignments identical.	Passed

3.3.3. Test Cases for Disruption Handling, Reallocation and Updated Schedule

Table 7: Test Cases for Reallocation

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-034	Owner submits emergency unavailability request	Select B01, Emergency Unavailable; enter reason; submit.	Request recorded as Pending with confirmation.	Pending request recorded.	Passed
UAT-035	Owner submits specific-date request	Select B01 and date range; enter reason; submit.	Pending request records requested range.	Pending request with range recorded.	Passed
UAT-036	Invalid request is refused	Enter reversed dates or omit reason; submit.	Request refused with validation message.	Request refused; message shown.	Passed
UAT-037	Authority sees pending requests	Open Disruption Requests as authority.	List includes bus, route, type/period, reason, and status.	All details listed.	Passed
UAT-038	Authority rejects request	Reject pending request; inspect schedule.	Status Rejected; schedule unchanged.	Rejected; schedule unchanged.	Passed
UAT-039	Authority accepts request	Accept pending request; inspect bus and schedule.	Status Accepted, bus unavailable, reallocation starts.	Accepted; bus unavailable; reallocation started.	Passed
UAT-040	Owner sees decision	Open My Requests as owner.	Request displays Accepted or Rejected.	Decision displayed.	Passed
UAT-041	Only authority can accept request	Send accept request using owner token.	Access denied; status unchanged.	Access denied; status unchanged.	Passed
UAT-042	Emergency request processes next 24 hours	Accept emergency request for B01; inspect affected trips.	Only B01 trips in next 24 hours are affected.	Only next-24-hour trips listed.	Passed
UAT-043	Past trips are not changed	Raise request after some turns have run; inspect schedule.	Completed/started turns retain bus, driver, and status.	Past turns unchanged.	Passed
UAT-044	Candidate with assignment conflict rejected	Give B02 overlapping assignment; process disruption.	B02 rejected and conflict reason recorded.	B02 rejected for conflict.	Passed
UAT-045	Candidate without turnaround rejected	Set previous trip to end less than 15 minutes before affected trip.	Candidate rejected for insufficient turnaround.	Candidate rejected for turnaround.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-046	Unavailable bus excluded	Keep B06 unavailable; process B01 disruption.	B06 excluded from candidates.	B06 excluded.	Passed
UAT-047	Daily workload breaks route-compatibility tie	Set B03 to 3 turns and B04 to 5, equal compatibility.	B03 ranked first and selected.	B03 selected.	Passed
UAT-048	Recent reallocations break workload tie	Prepare candidates equal on compatibility/workload but different recent reallocations.	Candidate with fewer recent reallocations selected.	Candidate with fewer reallocations selected.	Passed
UAT-049	Successful reallocation is stored	Process disruption with feasible bus and driver; inspect turn.	New bus and driver shown; status Reallocated.	New bus/driver and status stored.	Passed
UAT-050	Affected trips decided independently	Use B01 turns at 08:00, 10:00, 12:00, 14:00; report unavailability at 09:30.	10:00 cancelled for preparation time; 12:00 reallocated to B03/D03; 14:00 receives separate decision.	10:00 cancelled; 12:00 reallocated; 14:00 decided separately.	Passed
UAT-051	Authority portal shows updated schedule	Process request yielding reallocation and cancellation; inspect portal.	Both statuses are clearly marked.	Both statuses marked.	Passed
UAT-052	Decision stored for traceability	Open history for a reallocated and cancelled turn.	History records event, affected turn, old/new resources or cancellation, decision, and reason.	Required history details recorded.	Passed

3.4. Real-Time Driver Behavior Telematics and Stakeholder Portals

3.4.1. Test Scenario

This scenario validates safety-event detection and stakeholder views. During a trip, the test vehicle performs controlled maneuvers to cross or remain below configured thresholds. Tests assess overspeeding, aggressive braking, aggressive turning, abnormal slow movement, alert latency, false alerts from road roughness or congestion, event storage, safety-score updates, and role-specific access to safety information.

3.4.2. Test Cases for Bus

Table 8: Test Case for Driver Behavior

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-053	Overspeeding raises in-cab alert	Start trip as D01; drive above 60 km/h for over 5 s.	Audio and visual overspeeding alert appear.	Audio and visual alerts appeared.	Passed
UAT-054	Short speed burst raises no alert	Exceed 60 km/h for under 5 s; slow down.	No overspeeding alert.	No alert raised.	Passed
UAT-055	Speed within limit raises no alert	Drive at about 58 km/h for one minute.	No overspeeding alert.	No alert raised.	Passed
UAT-056	Aggressive braking raises alert	Brake so deceleration exceeds -3.5 m/s².	Aggressive-braking alert appears.	Alert appeared.	Passed
UAT-057	Normal braking raises no alert	Brake smoothly at about -2.0 m/s².	No braking alert.	No alert raised.	Passed
UAT-058	Aggressive turning raises alert	Turn sharply above 25 km/h with yaw rate above 0.45 rad/s.	Aggressive-turning alert appears.	Alert appeared.	Passed
UAT-059	Same yaw rate at low speed raises no alert	Turn at 25 km/h or less with yaw rate above 0.45 rad/s.	No turning alert.	No alert raised.	Passed
UAT-060	Abnormal slow movement raises alert	Drive below 10 km/h for 60 s on a clear road.	Abnormal slow-movement alert appears.	Alert appeared.	Passed
UAT-061	Slow movement in congestion raises no alert	Drive below 10 km/h for 60 s with congestion ahead.	No abnormal slow-movement alert.	No alert raised.	Passed
UAT-062	Alert displays details	Trigger an alert and inspect screen.	Event type, measured value, and time shown.	Details shown.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-063	Alert reaches driver within 1.5 seconds	Trigger alert; compare maneuver and alert timestamps.	Alert appears within 1.5 s.	Alert appeared within 1.5 s.	Passed
UAT-064	Road roughness does not trigger braking alert	Drive over rough surface at steady speed.	Vertical vibration does not trigger aggressive-braking alert.	No braking alert raised.	Passed
UAT-065	Each violation is stored	Trigger each alert type; inspect trip history.	Each record stores trip, type, value, time, and location.	All details stored.	Passed
UAT-066	Safety score drops after violation	Record D01 score; trigger violation; inspect score.	Score updates to reflect violation.	Score updated.	Passed
UAT-067	Smooth trip leaves score unchanged	Record D02 score; complete smooth trip; inspect score.	No alert; score unchanged.	No alert; score unchanged.	Passed
UAT-068	Driver sees own score and history	Log in as D01; open Safety Score.	Driver sees own score and violations only.	Own score and violations shown.	Passed
UAT-069	Driver cannot see another driver's data	As D01, request D02 safety data via Postman.	Access denied; no data returned.	Access denied; no data returned.	Passed
UAT-070	Owner sees safety summary of owned drivers	Open Driver Safety in owner application.	Safety summary for owned buses' drivers appears.	Summary shown.	Passed
UAT-071	Authority portal shows live safety events	Open Live Map; trigger alert on test vehicle.	Event appears against bus on live map.	Event appeared against bus.	Passed
UAT-072	Alert dispatch status is recorded	Trigger alert; inspect safety record through Postman.	Record contains alert ID and push status.	Alert ID and push status recorded.	Passed

3.5. Distance-Based Revenue Management

3.5.1. Test Scenario

This scenario validates the financial calculation using verified kilometres grouped by the time slot of the scheduled turn. The base rate is LKR 12.00/km, with multipliers of 1.30 for Peak, 1.00 for Normal, and 0.70 for Night. Eligible completed night trips on designated Nisi Seriya corridors receive an incentive of LKR 25.00 per verified kilometre, deducted from the gross permit fee. Tests cover individual rates, a mixed month, surplus credit, unverified distance, reallocated trips, role-based access, and settlement.

3.5.2. Test Cases for Distance-based Revenue Management

Table 9: Test case for Distance-Based Revenue Management

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-073	Owner views monthly bill	Log in; open My Bills; select September 2026.	Gross fee, night credit, and net due shown.	All three values shown.	Passed
UAT-074	Peak distance uses Peak rate	Prepare 100 verified Peak km; inspect bill.	Gross fee LKR 1,560.00 (100 × 12 × 1.30).	LKR 1,560.00.	Passed
UAT-075	Normal distance uses Normal rate	Prepare 100 verified Normal km; inspect bill.	Gross fee LKR 1,200.00.	LKR 1,200.00.	Passed
UAT-076	Night distance uses Night rate	Prepare 100 verified Night km; inspect bill.	Gross fee LKR 840.00.	LKR 840.00.	Passed
UAT-077	Mixed month calculated correctly	Prepare 100 Peak, 200 Normal, and 50 Night km, all verified on designated corridor.	Gross LKR 4,380.00; credit LKR 1,250.00; net LKR 3,130.00.	Values matched expected amounts.	Passed
UAT-078	Verified night trip receives incentive	Complete verified 50 km night trip on designated Nisi Seriya corridor.	Night credit LKR 1,250.00.	LKR 1,250.00.	Passed
UAT-079	No incentive outside designated corridors	Complete 50 km night trip on non-designated corridor.	Credit is LKR 0.00.	LKR 0.00.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-080	Incomplete night trip earns no incentive	End trip before completion criteria are met.	No night credit is applied.	No credit applied.	Passed
UAT-081	Unverified distance is not billed	Complete trip with 1 km deviation; inspect distance and bill.	Outside-corridor kilometres excluded from billable distance.	Outside-corridor distance excluded.	Passed
UAT-082	Rate follows scheduled turn's time slot	Run turn stamped Night; inspect bill line.	Night multiplier 0.70 applied.	Night multiplier applied.	Passed
UAT-083	Reallocated turn billed to bus that ran it	Reallocate turn to another operator's bus; run; inspect both bills.	Replacement operator is billed, not original operator.	Replacement operator billed.	Passed
UAT-084	Logged distance agrees with odometer	Drive known route; compare logged and odometer distances.	Agreement exceeds 98%.	Agreement exceeded 98%.	Passed
UAT-085	Owner cannot see another operator's bill	As B01 owner, request another operator's bill via Postman.	Access denied; no data returned.	Access denied; no data returned.	Passed
UAT-086	Authority views any operator's bill	Log in with finance role; select operator and month.	Selected operator's bill is shown.	Bill shown.	Passed
UAT-087	Billing cycle is settled	Settle October 2026 cycle for an operator.	Statement shows gross fee, verified km, night credit, and net balance.	All four values shown.	Passed
UAT-088	Unknown-operator settlement refused	Settle a nonexistent operator.	Settlement refused with operator-not-found message.	Settlement refused; message shown.	Passed
UAT-089	Settlement with unverified distance refused	Select cycle containing unverified distance; settle.	Settlement refused with an unverified-distance message.	Settlement refused; message shown.	Passed

3.6. End-to-End Integration

This scenario validates the closed loop across the four components. It tests a scheduled trip from generation through telemetry and billing, the propagation of route deviations into portal alerts and audit records, the distribution of safety events to stakeholders, disruption-driven reallocation and subsequent tracking, night tariffs and incentives, buffered telemetry after network loss, and refusal of insecure MQTT connections.

Table 10: Test Case for End-to-End Integration

ID	Description	Test Steps	Expected Result	Actual Result	Status
UAT-90	Complete trip from schedule to bill	Generate schedule; start assigned turn as D01; drive Route 138 with telemetry; end trip; inspect owner bill.	Trip linked to turn; distance verified and billed at turn's time-slot rate.	Trip linked and distance billed at slot rate.	Passed
UAT-91	Deviation reaches portal, audit, and bill	Leave corridor for over 30 s; inspect alerts, audit, and bill.	Alert and audit record appear; outside-corridor distance not billed.	Alert/audit shown; distance excluded.	Passed
UAT-92	Unsafe maneuver reaches all roles	Brake beyond -3.5 m/s²; inspect driver, owner, and authority views.	Driver receives alert; owner and authority see event against trip.	Alert and event visible to all roles.	Passed
UAT-93	Disruption leads to tracked reallocated turn	Owner requests B01 unavailability; authority accepts; D03 receives alert and starts turn; replacement bus sends telemetry.	Turn reallocated; driver notified; telemetry tied to reallocated turn.	Reallocation, notification, and telemetry linkage confirmed.	Passed
UAT-94	Verified night trip earns night rate and credit	Run night turn on designated Nisi Seriya corridor; inspect bill.	Night multiplier 0.70 applied and LKR 25/km	Night rate and credit applied.	Passed

ID	Description	Test Steps	Expected Result	Actual Result	Status
			credit deducted.		
UAT-95	Buffered telemetry included in bill	Disable cellular connection during trip; restore; end trip; inspect bill.	Buffered records processed and distance included in verified total and bill.	Buffered distance included.	Passed
UAT-96	Connection without TLS is refused	Attempt MQTT without TLS and with invalid device credentials.	Both attempts refused; no telemetry stored.	Both attempts refused; no telemetry stored.	Passed

4. Conclusion

The User Acceptance Testing of BusSync indicates that the four components and their connections met the specified functional, algorithmic, usability, and security criteria in the controlled test environment. Across six scenarios and 96 test cases, the platform was exercised through the authority web portal, owner and driver mobile applications, REST API, and an ESP32 edge unit mounted in a test vehicle. Coverage included authentication and access control; bus tracking, route violation and halt detection; monthly scheduling, time-slot management and disruption handling; distance-based revenue management; driver-behavior telematics and stakeholder portals; and end-to-end integration.

The recorded results indicate that telemetry reached the backend within the stated latency target, was recovered after GPS or network loss, and supported verified-distance calculations that excluded travel outside the corridor. Route deviations were flagged after the configured persistence period, and halts were detected. The monthly base schedule contained the expected assignments, distributed work evenly, avoided overlaps and turnaround breaches, and labelled turns with time slots. Disruption events resulted in recorded reallocations or cancellations with reasons. Bills followed the configured tariff and night-incentive rules, including the tested surplus case. Driver alerts were reported within 1.5 seconds under the test conditions, and role-based access restrictions were enforced.

Overall, the recorded UAT results support the conclusion that BusSync satisfied the tested acceptance criteria in the controlled environment. Deployment readiness remains subject to final refinements, institutional approval, and confirmation that the reported test records and measurements are retained as evidence. As the decision engines are rule-based and algorithm-based, acceptance was assessed by comparing each output with the expected result for the prepared test data; all 96 cases are recorded as passed.